

Shrisha Hegde

Python Full Stack Developer

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Summary

I am a passionate Python Full Stack Developer with 3.2 years of professional experience in designing, developing, and deploying scalable web applications using Python frameworks and Vue.js. I have a strong background in microservices architecture, serverless systems, and AI/ML-integrated solutions including RAG architecture, semantic search, and real-time audio processing. My expertise spans RESTful API design, cloud infrastructure (AWS), and building secure, high-performance systems. I thrive in Agile team environments where collaboration, innovation, and quality drive project success.

Skills

- **Languages & Frameworks:** Python, JavaScript, TypeScript, SQL, FastAPI, Vue.js (Composition API, Pinia, Vue Router)
- **AI/ML & Data:** RAG Architecture, Semantic Search, OpenSearch k-NN, AWS Bedrock, NLP, Prompt Engineering
- **Cloud & DevOps:** AWS (Lambda, ECS Fargate, SNS/SQS, DynamoDB, API Gateway, S3, EventBridge, CDK), Terraform, Docker, GitLab CI/CD
- **Frontend & UI:** HTML, CSS, Tailwind CSS, Vuetify, D3.js, Chart.js, Ag Grid, WebSockets
- **Security & Testing:** JWT, OAuth2/OIDC, RBAC, AWS Secrets Manager, SlowAPI, pytest, Postman, Swagger
- **Tools & Practices:** Git, Linux, Agile (Scrum, Jira), Microservices Architecture, IaC

Experience

Mercedes-Benz Research & Development India (MBRDI) - Software Development Engineer August 2023 – Present

- Built responsive and modular frontend interfaces using **Vue 3** Composition API, **Pinia** state management, and **TypeScript** for internal enterprise applications.
- Developed production REST APIs using **FastAPI** with async/await patterns, **Pydantic** validation, and dependency injection.
- Architected serverless event-driven systems using **AWS Lambda**, **SNS/SQS**, **DynamoDB**, and **API Gateway** following microservices architecture.
- Provisioned cloud infrastructure using **AWS CDK**, **Terraform**, and built **GitLab CI/CD** pipelines for multi-environment deployments.
- Implemented security across services using **OAuth2/OIDC**, **JWT**, **RBAC**, and **AWS Secrets Manager**.
- Designed **RAG** architecture with semantic search and integrated **AWS Bedrock** for AI-powered recommendation and interview automation platforms.
- Built real-time audio pipelines using **AWS Transcribe** and **Polly** with **WebSocket** communication for voice-based AI interactions.

Mercedes-Benz Research & Development India (MBRDI) - Intern January 2023 - July 2023

- Gained hands-on expertise in **Vue.js**, **JavaScript**, **TypeScript**, and data visualization libraries like **D3.js** and **Chart.js** while single-handedly building the complete UI for an internal application.
- Adopted best practices in component-driven development, containerized deployments with **Docker**, and role-based access control patterns.

Projects

Live Map Platform

Vue.js, TypeScript, D3.js, Chart.js, Ag Grid, Docker

An internal platform to visualize error logs and manage releases across markets, with two use-cases built entirely from scratch.

- Implemented a release management service to track software releases across deployment stages and regions, using **Ag Grid** for tabular data views.
- Built an error handling service with an interactive dashboard to visualize live streaming system errors in multiple formats using **D3.js** and **Chart.js**.
- Integrated **TypeScript**, **Pinia** for state management, and **Vue.js Composition API** for scalable component development.
- Implemented role-based access control (RBAC) and containerized the application using **Docker**.

Team Onboarding Platform

Python 3.12, FastAPI, AWS Lambda, SNS/SQS, DynamoDB, AWS CDK, GitLab CI/CD, Docker

A serverless event-driven platform using AWS Lambda (11+ functions) to automate team onboarding across 8+ DevOps tools and multi-cloud infrastructure via a single API.

- Built serverless microservices using **AWS Lambda** and **API Gateway** with fault-tolerant parallel execution, automatic retries, and dead-letter queue recovery.
- Reduced manual team onboarding from 3-5 days to under 10 minutes by automating provisioning across all tools with parallel execution, eliminating human error and multi-admin coordination.
- Developed REST APIs using **FastAPI** with `async/await`, **Pydantic** validation, dependency injection, and auto-generated OpenAPI documentation.
- Developed the frontend using **JavaScript** and **Vue.js** Composition API for a responsive and modular user interface.
- Set up **SNS** fan-out to multiple **SQS** queues for parallel Lambda invocations with exponential backoff retries, and implemented GitLab webhook integration for automated pipeline event processing and status tracking.
- Used **DynamoDB** for centralized request tracking with real-time status updates, audit trails, and TTL-based auto-expiration.
- Implemented **JWT** authentication with API Gateway authorizers and **OAuth2** flow with role-based access control across DEV/NONPROD/PROD environments.
- Provisioned all infrastructure using **AWS CDK** (Python) — Lambda functions, DynamoDB tables, SNS/SQS, IAM roles — across 3 AWS accounts, with secrets managed via **AWS Secrets Manager**.
- Built **GitLab CI/CD** pipeline with multi-environment deployment and manual approval gates across DEV, NONPROD, and PROD.

AI Interview Platform

FastAPI, Vue 3, AWS Bedrock, Transcribe, Polly, Terraform, ECS Fargate, PostgreSQL

A full-stack recruitment automation platform that uses AI for dynamic interview question generation, real-time voice interactions, and intelligent candidate evaluation.

- Built the platform using **FastAPI** (async Python) and **Vue 3**, integrating **AWS Bedrock** (Claude 3 Sonnet) for dynamic question generation and context-aware interview evaluation.
- Implemented real-time audio processing using **AWS Transcribe** Streaming with **WebSocket** communication for speech-to-text and **AWS Polly** neural TTS for natural voice synthesis.
- Built a resume matching engine using **AWS Bedrock** for semantic analysis of job descriptions and candidate profiles with multi-factor scoring to surface top candidates.
- Designed AWS infrastructure with **Terraform** deploying **ECS Fargate** containers, **RDS PostgreSQL** with async **SQLAlchemy**, **ALB**, and **S3** for audio storage.
- Implemented security with **SlowAPI** rate limiting, PII redaction using **NLP**, **OAuth**-based authentication with **RBAC**, and security headers (CSP, HSTS).
- Accelerated development by leveraging AI-assisted code generation tools and **prompt engineering** for rapid prototyping and implementation across the full stack.

AI Learning Recommendation System

FastAPI, Vue 3, TypeScript, OpenSearch, AWS ECS Fargate, Terraform, DynamoDB

An AI-powered course recommendation platform serving personalized suggestions from 20K+ courses to 50K+ employees, reducing course discovery time from hours to seconds.

- Designed a **RAG** architecture with a two-stage semantic search engine using text embeddings and cross-encoder reranking on **Amazon OpenSearch** k-NN (HNSW), solving the accuracy-cost tradeoff by achieving sub-150ms inference on CPU-only infrastructure — reducing costs by 90% compared to GPU-based approaches.
- Designed an event-driven pipeline using **AWS EventBridge**, **S3**, and **ECS Fargate** to automate course indexing and IDP processing, reducing manual workflow from 8 hours to 15 minutes with **Fernet** encryption for PII protection and department-based vendor access controls.
- Deployed production infrastructure using **Terraform** orchestrating **ECS Fargate**, **DynamoDB**, **OpenSearch**, and **ALB**, achieving 99.9% uptime for enterprise-scale recommendations.
- Developed full-stack application with **Vue 3 + TypeScript** frontend and **FastAPI** backend, implementing **OAuth2/OIDC** authentication with Mercedes-Benz SSO and role-based access via HttpOnly JWT cookies.

Education

